Predicting Clothing Purchase Behaviour Using Transaction and Product Data in Online Fashion Retail

MSc Research Project
Masters in Artificial Intelligence

Priyanka Bellamkonda
Student ID: 23338849

School of Computing
National College of Ireland

Supervisor: Mohit Garg

National College of Ireland

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School of Computing

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Student ID:	23338849 ……
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Predicting Clothing Purchase Behaviour Using Transaction and Product Data in Online Fashion Retail

Priyanka Bellamkonda
23338849

Abstract

This research examines the effectiveness of machine learning models in forecasting customer buying behavior in online fashion retail through transactional, product, and customer data. The study utilizes the H&M Personalized Fashion Recommendation dataset available on Kaggle, which includes more than 31 million transactions, 100,000 products, and 1.3 million customers. A dataset with four million records was created using actual purchases as positive examples and realistically generated non-purchases as negative examples to avoid data leakage and artificial enhancement of model performance. Two models were created and assessed: XGBoost, a tree-based ensemble method, and TabNet, a deep learning framework tailored for tabular data utilizing attention-driven feature selection. Both models were trained using the same experimental setup and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost achieved an accuracy of 76.65% and a ROC-AUC of 0.8455, while TabNet achieved 77.74% accuracy and a ROC-AUC of 0.8584, showing a modest but consistent improvement at higher computational cost. Customer segmentation using K-means clustering demonstrated that TabNet's performance gains were consistent across all demographic groups with no significant bias detected. A business value analysis indicated that TabNet's 1.09 percentage point enhancement could potentially yield an estimated EUR 5.1 million in extra annual profit on a large scale. The research shows that contemporary machine learning models, when backed by realistic data generation and thorough evaluation, can effectively and cost-efficiently forecast purchasing behavior in extensive online fashion retail settings.

Keywords: Online Fashion Retail, Purchase Prediction, Machine Learning, XGBoost, TabNet, Business Value, ROI

1 Introduction

Motivation

The fashion retail business has been dramatically changing digitally in the last decade as online portals have come to be the main platform that allows them to view, engage with, and shop. The change has helped retailers to amass great amounts of transactional, product and customer data that are produced as a result of digital interactions. Using this type of data to learn and anticipate customer buying behavior has taken the center stage in enhancing inventory control, personalization policies and marketing performance. Consequently, the data-driven decision-making currently becomes one of the primary sources of competitive advantages in online fashion retail.

Predictive modelling is very important in ensuring that the retailers predict the purchasing behavior of customers based on the interactions in the past. Machine learning applications can be run to assist most retail applications, such as customized recommendation engines, demand management, and inventory optimization, by recognizing trends in customer behaviors and product characteristics. Earlier studies in this field have been mainly based on collaborative filtering, rule-based systems and classical models of machine learning including logistic regression, support machine and decision trees. Although these methods have been found to perform reasonably, they can be associated with the difficulty of scaling, generalization, and interpretability especially when it comes to large commercial datasets with diverse types of features.

Later developments have proposed ensemble algorithms and neural networks as a solution to these issues. Nevertheless, the biggest approach methodology problem with purchase prediction studies is the formation of negative samples. Most experiments lead to non-purchase occurrences when customer-item pairings are randomly formed without any realistic transactional characteristics, i.e. price or channel-of-sale information. This habit may create data leakage and any model may take advantage to learn artificial patterns and not acquire authentic purchase behavior, which will inflate performance measurements reported. This is a critical matter but it is not explicitly discussed in the literature especially in giant scale fashion retail settings.

Thesis Approach

The proposed MSc research project aims at forecasting customer purchase likelihood based on structured transactional, product, and customer data of the H&M Personalized Fashion Recommendation Dataset. The data that is publicly available on Kaggle through H&M has more than 30 million anonymized purchase orders, descriptive product (type, department, color, and garment group) and customer profile details. Both the size and the complexity of this data are both an opportunity and a challenge: on the one hand, it is a realistic means of modelling customer behavior; on the other, to extract meaningful insights, it is necessary, both to have a robust preprocessing of the data and to have scalable techniques of modelling.

In order to overcome these difficulties, this report compares two machine learning methods that deal with structured data. The former is the XGBoost, which is a popular gradient boosting algorithm with good performance and efficiency with tabular data. The second one is TabNet which is a deep learning system designed specifically to handle tabular data that uses sequential attention and selective sparse features to enhance predictive performance and interpretability. Through the comparison of these models in the same conditions of an experiment, this report explores whether any practical advantages can be gained using modern deep learning architecture over the time-tested ensemble methods when it comes to fashion purchase prediction. Moreover, the K-means clustering method is used to cluster customers depending on age, club membership, and the number of times they receive fashion news to analyze whether there is a difference in predictive performance among the demographical groups.

Objectives

The main aim of the study is to investigate how the data on transactional and product attributes can be employed to model customer purchase behavior on a large scale. In particular, this report conducts a comparison and analysis of XGBoost and TabNet based on conventional classification metrics, as well as the interpretability, cost of computation, and practical deployment factors. Through a realistic negative sampling, model comparison, and business-focused evaluation, this report will allow closing the gap between the methodological rigor and practical applicability in fashion retail analytics.

Therefore, the following question will direct the research:

How effectively can machine learning models predict customer purchase behavior in the online fashion industry using transactional and product attribute data?

Thesis Scope

The rest of the report is structured in the following way. In section 2, related work on purchase prediction and machine learning techniques on structured data is reviewed. The research methodology is outlined in Section 3. Section 4 describes the system and model design. The process of implementation is described in Section 5. Section 6 shows the analysis of the evaluation results and comparison. Lastly, Section 7 provides the conclusion to the report and limitations of future research.

2 Related Work

In this section, I review key studies and methodologies relevant to purchase prediction in e-commerce, machine learning applications in retail, and recent deep learning techniques for tabular data. The review is organized in two subsections: (1) classical and ensemble ML models in retail, and (2) deep learning methods for tabular data. I conclude by highlighting their strengths, limitations, and the gap my work fills.

2.1 Classical and Ensemble Models in E-commerce Purchase Prediction

Many studies on e-commerce purchase prediction have adopted traditional and ensemble machine learning models. For example, (Wang et al., 2025) used a Random Forest classifier on e-commerce behavior data to predict consumer purchases and reported solid accuracy, though they noted limitations in capturing temporal dependencies. Other researchers combine boosting approaches such as XGBoost, LightGBM, and random forests to build ensemble models, arguing that such combinations yield superior robustness and generalization over single models. In a logistic regression baseline study, (Balyemah et al., 2024) applied logistic regression and compared it with random forest in predicting purchase behaviour on online platforms, evaluating metrics such as precision, recall, and F1 score. Their conclusions emphasized interpretability but noted reduced capacity when feature relationships are non-linear or complex.

(Ran et al., 2024) proposed a hybrid framework combining temporal attention encoding, graph neural networks, and meta-learning to enhance purchase prediction, showing

gains over classical ML baselines. Their approach underscores the value of integrating structural relationships and temporal dynamics in prediction tasks.

(Esmeli & Gokce, 2025), studied post-cart behavior, incorporating session-specific factors (price variation, product category transitions, geographic context) to improve prediction of purchase after cart addition. They used multiple ML models and achieved high F1 scores, showing the value of session-level features in purchase modelling. While classical and ensemble methods remain strong baselines, they face limitations: (a) they often require extensive feature engineering, (b) they may struggle to model temporal or sequential dependencies in user behavior data, and (c) model interpretability can degrade when combining many components.

2.2 Deep Learning for Tabular Data & Interpretability Methods

In recent years, researchers have strived to bridge the gap between deep learning and tabular data, seeking methods that match or exceed tree-based model performance while offering representation learning and interpretability.

TabNet, as discussed by (Arik & Pfister, 2020), is a pivotal architecture designed specifically for tabular data. It employs sequential attention and sparse feature selection masks to dynamically choose which features to use at each decision step, thereby allowing interpretability and efficient learning. Their experiments, (Arik & Pfister, 2020), showed that TabNet can outperform conventional neural nets and compete with tree models across various benchmark datasets. (Arik & Pfister, 2021) Critics, however, have raised concerns about performance and training speed: some practitioners report that well-tuned XGBoost still outperforms TabNet on many real-world tabular datasets and that TabNet's training is slower and more resource-intensive.

Other architectures have extended the deep learning paradigm on tabular data. The TabTransformer model, introduced in 2020, applies Transformer layers to categorical feature embeddings, enhancing feature interactions and handling missing data robustly (Huang et al., 2020). It demonstrates competitive performance with tree models, especially when categorical features are dominant.

The broader taxonomy of deep tabular learning methods includes architectures such as SAINT, ARM-Net, NODE, and FT-Transformer, which leverage attention, hybrid learning, or differentiable tree structures. For instance, newer work like InterpreTabNet (2023) builds on TabNet to improve interpretability and stabilization of feature importance attributions (Wa et al., 2023). These newer variants attempt to reduce the "black box" criticism while maintaining competitive accuracy.

Strengths of deep tabular methods include:

- End-to-end learning without handcrafted features
- Ability to learn representations across mixed feature types
- Intrinsic interpretability when attention or masking is used

Weaknesses include:

- High training and tuning complexity
- Larger computational cost
- Sensitivity to hyperparameter selection
- In many practical datasets, gains over tree-based models are modest or inconsistent

2.3 Synthesis, Gaps, and Justification for This Study

An analysis of available sources demonstrates that there are some significant gaps in the current methods of purchase prediction in e-commerce and fashion retail. Logistic regression, Random Forests and XGBoost are classical and ensemble machine learning models that are still used predominantly because of their effectiveness and strength on structured data. Nonetheless, these approaches tend to rely on a lot of feature engineering, and cannot model complex interaction of features with meaningful tuning. Also, the majority of studies center around generic e-commerce data where little emphasis is put on the unique aspects of fashion retail where the outcomes of the purchasing behavior highly depend on the product properties, consumer tastes and interaction patterns.

A certain methodological gap is a critical approach to negative samples in purchase prediction exercises. Most of the studies that have been conducted in the past provide non-buy instances by utilizing random customer products combinations without providing realistic transactional attributes like price or channel of sale. This practice will cause data leakage whereby models will take advantage of missing or unrealistic feature values instead of learning true behavioral patterns. Therefore, performance measures as reported can be far above actual predictive ability. Though it is important, the issue has not been explicitly discussed in the current literature, especially in massive commercial data.

The recent deep learning architecture, which has been trained on tabular data, like TabNet and Tab Transformer, have proven to have some benefits in representation learning and interpretability with respect to attention mechanisms. Nevertheless, there is little empirical analysis of these models applied in large scale fashion retailing. There are relatively a limited number of studies that give controlled comparisons between deep tabular and existing ensemble algorithms on identical preprocessing pipelines with respect to whether the extra complexity brings about significant practical advantages. Moreover, most of the assessments have been based exclusively on aggregate accuracy without consideration of computational cost and deployment.

The other weakness of the literature is that it has not analyzed performance of models in terms of customer demographics. Majority of the studies describe general measures without analyzing whether predictive accuracy is dependent on age, loyalty status or level of engagement and thus the study is not able to provide insight into strength and possible bias.

The report provides solutions to these gaps by introducing realistic negative sampling, controlled comparison of XGBoost and TabNet on a large-scale fashion data, and measuring consistency of the performance of the models within customer segments. The combination of

these contributions will give a more valid, interpretable, and practically relevant evaluation of the purchase prediction models within the online fashion retail.

3 Research Methodology

This section summarizes the methodology applied to assess predictive accuracy of machine learning models at predicting customer purchase behaviour in online fashion retailing. The methodology is developed to have transparency, reproducibility and fairness in comparison of models. It addresses the description of the dataset, data preparation and sampling plan, feature engineering, model choice and configuration, client segmentation, business worth modelling, and metrics of evaluation. Every decision taken about the methodology is based on the existing research on the topic of machine learning and previous experiments with predicting purchases and modelling tabular data.

3.1 Dataset Description

The data applied in this study is the H&M Personalized Fashion Recommendation Dataset which is openly accessible to H&M and is located at the Kaggle platform. It includes 3 main files, including transactionstrain.csv, more than 31 million records anonymised customer purchases; articles.csv, which includes the metadata of over 100,000 fashion products, and customers.csv, which contains both demographic and engagement data of about 1.3 million customers. The data set is two years of September 2018 to September 2020 and represents online retail in the real world on the enterprise scale.

The transaction data contains the purchase timestamps, product identifiers, prices, and sales channel information whereas the product metadata consists of such attributes as product type, department, colour group, index, section, and garment group. The customer level information entails age, membership status of the club, and overall frequency of involvement in the fashion news. Collectively, these datasets offer a deep picture of the interactions between customers and products which can be used in the modelling of purchase behaviour using structured data.

3.2 Experimental Environment and Tools

All the tests were written in Python 3.10 in a Google Colab Pro+ in the environment with a GPU acceleration. The choice of this setup was based on the fact that it enables effective processing of large-scale datasets and training of deep learning models. Other libraries that are core are Pandas and NumPy to manipulate the data, Scikit-learn to pre-process and evaluate, XGBoost to implement a gradient boosting model, and PyTorch TabNet to implement a deep learning model on tabular data. Exploratory analysis and visualization were done using Matplotlib and Seaborn. A fixed random seed (RANDOMSTATE = 42) was used to guarantee reproducibility; it was used throughout the data sampling, model training, and the clustering processes.

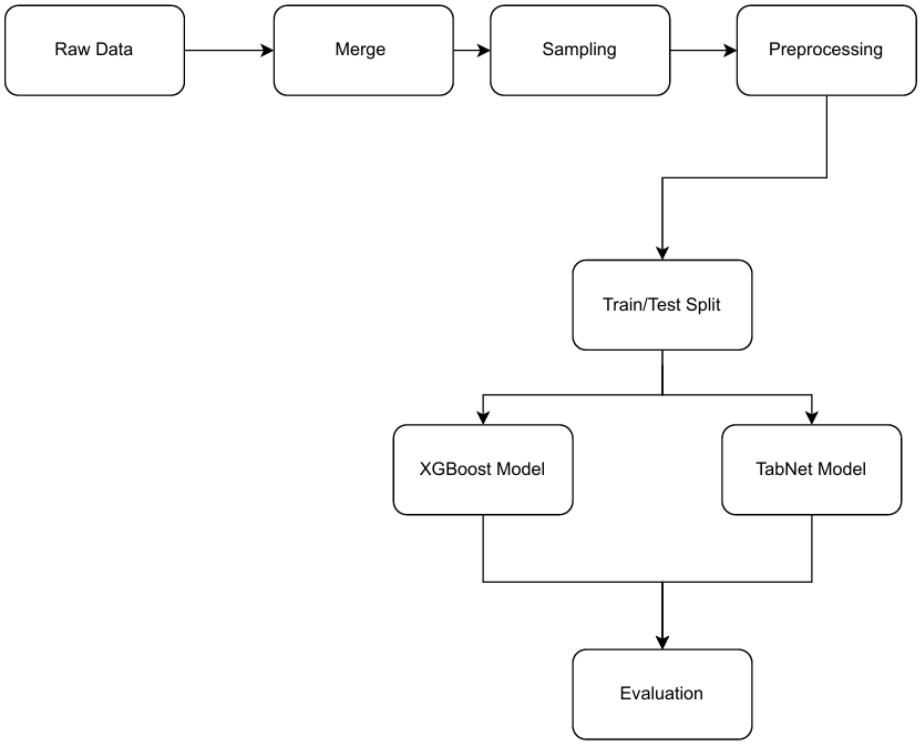

Figure 1 : Project Pipeline

3.3 Data Sample and Class Construction.

Since the size of the original transaction log is large, a random sample size of two million actual purchase transactions was obtained to form the positive class. All transactions are observed purchase events and the foundation of purchase behavior. These records were used to combine customerid and Articleid keys with customer and product metadata to form a combined set of records comprising of transactional, product, and customer attributes.

The negative class construction was considered as a methodological step of the critical kind. The non-purchase cases were created at random customer-product pairings which failed to occur in the initial transaction record. Instead of abandoning these synthetic samples with incomplete transactional characteristics, however, natural price and sales channel values were attached by sample of the empirical distributions that were observed in the positive class. This is used to reduce leakage of data because models cannot use the missing or unrealistic values of features to differentiate between classes.

After this procedure, the dataset was balanced such that it had four million records, including two million purchase records and two million non-purchase records. Such a balanced design helps to properly evaluate the classification performance and not to be biased towards the majority class.

3.4 Data Preprocessing and Engineering.

The preprocessing measures were used to guarantee the quality of data and compatibility of models. Numeric variables including the age of customers had missing values that were also imputed by using the median in order to reduce the effect of outliers.

Dichotomous or categorical variables such as club membership status and fashion news frequency were filled in with mode.

Twelve features were chosen to be modelled, which included transactional, product-level and customer-level information. Price and sales channel identifier were used as transactional features. The metadata of the product included the name of product type, product group name, color group name, department name, index name, section name and garment group name. Attributes of customers were age, member of the club and frequency of fashion news. All the categorical variables were coded with the help of label encoding, which allowed their application in both XGBoost and TabNet without extra embedding layers.

3.5 Model Choice and Model Set-up.

There were two chosen predictive models to reflect opposite approaches toward the structured data modelling. XGBoost has been selected as a good ensemble baseline because of its popularity, efficiency and its past record of success on tabular data. TabNet was chosen because it is a deep learning model that is specifically targeted at tabular data, which has a feature selection based on attention and a natural interpretation.

The same stratified 80:20 train-test split was used to train both models so as to give consistency and comparability. XGBoost was set by using 100 estimators, the learning rate was 0.1, the maximum tree depth was 6, subsample and column sample rates were 0.8. These parameters are representative of trade-offs that represent commonly used defaults and performance and computational efficiency.

TabNet was trained on a batch size of 1024 and Adam optimizer with learning rate of 0.02. The model parameters on capacity were adjusted to nd = 64, na = 64 and gamma = 1.5. A patience of 10 epochs was used to early stop and limit the training time. Both the models were trained on the same sets of features and target labels.

3.6 Customer Segmentation through Clustering.

The K-means clustering was applied on the test set to segment the customers into different groupings in order to test the hypothesis that predictive performance differed among demographic groups. Three attributes in the form of customers such as age, club membership status and frequency of fashion news were chosen as clustering features since they were relevant to the purchasing behaviour and were found in all the records.

The K-means algorithm was first initialised by using k-means++ method and run in ten initialisations in order to enhance stability. The value of k was chosen to be 5 to trade off interpretability and segment granularity. Descriptive statistics, such as mean age, modal engagement characteristics, and relative cluster size in each of the clusters, were used to profile each cluster. The performance of every model was then rated independently in each cluster to determine robustness and demographic uniformity.

3.7 Business Value Quantification.

In addition to technical assessment, in this research, a business value model is used to generate a financial contribution to predictive performance gains. The baseline assumptions

were based on the industry standard benchmarks and adjusted to the H&M dataset setting. These are, client base of one point three million, an average of ten customised recommendations per customer per month, average purchase price of EUR 50, conversion ratio of 15 percent and a profit margin of 40 percent.

The linear increase in the accuracy of the model was multiplied by the number of recommendations per year to determine the number of extra correct decisions. This was then transformed into estimated annual profit gains. The cost of training was calculated by utilizing the information on the use of the GPUs and the cloud prices and then the ratios on the return-on-investment. Every assumption is clearly provided besides making it transparent and reproducible.

3.8 Metrics of Evaluation and Analysis.

The standard classification metrics, such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC) were used to evaluate model performance. These measures have been calculated with the help of Scikit-learn and compared with the test set that was held out. The ROC curves were generated using probabilistic predictions and the confusion matrices aided in a thorough error analysis.

Strict result tables and visualisations were used to make performance comparisons between XGBoost and TabNet. In comparing the two models in the same conditions, this methodology allows objective ranking to be done on trade-offs in terms of predictive accuracy, cost of computation, and interpretability.

4 Design Specification

This project is designed in the form of a modular architecture which allows to perform scalable data processing, model development and evaluation. The general working process is based on a linear data flow that starts with the raw data ingestion until preprocessing, feature engineering, and model training, as well as final prediction output.

Preprocessing architecture starts with the loading of the three main core data sets namely transactions, articles and customers. Articleid and customerid keys are used to combine these in order to create a single dataset that has both transactional behaviour and user-product context. Based on this combined data a positive class is formed on the basis of 2 million real transactions. An artificial negative sample is then created by selecting random (customer, article) pairs that are not included in the transaction log and realistic transactional data (price and saleschannelid) is added to avoid data leakage. Missing values are filled in with median or mode and categorical features filled in with LabelEncoder. The last dataset is divided into training and the test sets by means of stratified sampling that maintains balance of the classes.

The TabNet model used in the research is a deep learning model that works with tabular data. It combines the sequential attention and sparse feature selection. The model takes the features as input and works through several steps of decision-making, which are informed by learned masks that dictate the features that need to be focused on. These masks enhance better interpretability and efficient gradient propagation. TabNet is also capable of

end-to-end learning that does not require manual feature engineering and also the model gives feature importance scores and the class probabilities.

Contrarily, XGBoost is a tree-based gradient boosting model and constitutes ensembles of decision trees through sequential optimization. The tree tries to fix the errors of the last tree, which enables the model to learn non-linear, multifaceted interactions between features. XGBoost is incredibly fast, does not even bother with missing values, and is optimally designed to work with structured data.

The software in this project consists of Python 3.10, Google Colab Pro+ with the use of a graphics card, and a number of fundamental libraries: Pandas and NumPy to manipulate data, Scikit-learn to preprocess data and metrics, PyTorch TabNet to implement the model, and XGBoost to boost the gradient. It is a toolchain offering a versatile and reproducible platform to the full machine learning pipeline, including raw data to actionable model outputs.

5 Implementation

The next step of the implementation was to convert the cleaned and preprocessed dataset into the format that can be used by machine learning models and run the training and evaluation pipeline of both XGBoost and TabNet. This step commenced when all the exploratory data analysis, merging, balancing and encoding processes were done. Here the size of the dataset was 4 million rows, balanced (50/50) between purchase and non-purchase classes with 12 fully encoded input features and a target label binary.

The final dataset after transformation was stratified into a training and test dataset using stratified sampling in order to have equal proportions of classes. The input features and target labels were divided, reshaped and held in memory efficient arrays to enhance compatibility with both the models. Categorical values were already represented in numbers and could be used with TabNet, which does not need embeddings, and XGBoost which is able to accept integer inputs naturally. Neither normalization nor scaling was necessary and both of the models are not sensitive to the scaling of features.

The implementation of the model was done in two different branches of the notebook. With XGBoost, the default settings were used to initialize the model and then used on the training set with early stopping on the validation loss. After the training, the model was applied to produce a prediction on the test set, both class probabilities and hard labels. The results of the classification were measured by a confusion matrix, classification report, and ROC curve. The resulting XGBoost model was then saved with joblib in order to be inferred or deployed again in the future.

PyTorch TabNet library was used to implement a TabNet model. The architecture was set up with five decision steps, 64 decision and attention dimensions, sparse regularization and batch normalization. The maximum number of epochs that were trained was 10 and early stopping was used. The output consisted of attention masks, loss curves, predicted probabilities and the final classification label of the test set. The metrics used in evaluation were calculated like XGBoost and the model was stored with Checkpointing functionality of PyTorch.

Code was developed in Python and run in a Google Colab Pro+ setting with the training of TabNet through the use of a GPU accelerator. Pandas, numpy, matplotlib, scikit-learn, XGBoost and pytorch-tabnet were considered core libraries. The result of the implementation stage was that there were fully trained models, summary of classification performance, ROC curves and saved artifacts of both models available to be evaluated and compared.

6 Evaluation

This section is a detailed analysis of the trained models on a balanced test set. The objective is to determine their prediction ability and compare their strengths in regard to accuracy, robustness and interpretability. The assessment is based on the multiple statistical measures and visualizing methods that are used to justify the analytical research of the results.

6.1 Experiment / Case Study 1: XGBoost Model Evaluation

Initially, the XGBoost classifier with the training set of 3.2 million samples and the evaluation set of 800,000 balanced examples was used in the first experiment. The output of the model was estimated in the following measures: accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost was able to withstand generalization and did not overfit. Table 1 presents the classification performance of the XGBoost model on the training and test datasets using standard evaluation metrics.

Table 1: XGBoost results

Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Training Set	0.7666	0.7370	0.8290	0.7803	0.8456
Test Set	0.7665	0.7369	0.8290	0.7802	0.8455

XGBoost had a test accuracy of 76.65, precision of 73.69, recall of 82.90, F1-score of 0.7802 and ROC-AUC of 0.8455. The model exhibited high generalization and low overfitting was exhibited by the almost similar training and test set performance. Its high recall implies that it is effective to capture real behavior of purchase and the balanced F1-score represents a good balance between the precision and recall. The overall process of training took 51 seconds, which is indicative of the XGBoost accuracy with large-scale structured data.

6.2 Experiment / Case Study 2: TabNet Model Evaluation

In the second experiment, the TabNet was trained on the same dataset and its performance was tested upon completion of 10 training epochs. TabNet also takes tabular

features through attention masks and decision steps as opposed to XGBoost. This model was trained with the use of GPU acceleration because of the computing requirements. Table 2 summarizes the training progression of the TabNet model, showing validation accuracy and training loss across epochs.

Table 2: TabNet results table

Epoch	Training Loss	Validation Accuracy
0	0.6122	0.6786
1	0.6008	0.6831
2	0.6081	0.6670
3	0.6061	0.6704
4	0.5877	0.7179
5	0.5455	0.7409
6	0.5156	0.7476
7	0.4968	0.7682
8	0.4840	0.7718
9	0.4759	**0.7774**

TabNet (10 epochs with early stopping, trained with GPUs) was found to have a test accuracy of 77.74, a precision of 75.48, a recall of 82.17, a F1-score of 0.7869 and ROC-AUC of 0.8584. The model showed a slightly better performance on the majority of metrics especially in precision and AUC indicating that it discriminates between purchase and non-purchase classes better. This was however at a high computational cost since it took 11,095 seconds (about 3.1 hours) to train- 218-fold increase over XGBoost. TabNet has the advantage of being interpretable via the attention mechanism that allows the identification of the features that are paid attention to at each decision step, but this complexity also directly leads to longer training time.

Although TabNet has lower training time, its attention mechanism does have one more benefit, namely, interpretability, showing what features have been the most attended at each step of the decision-making.

6.3 Experiment / Case Study 3: Comparative Model Analysis

The two models were juxtaposed in this case study in terms of their performance measures, training time, and interpretability. The table below provides the summary of the comparative results:

Table 2: Model Comparison (XGBoost vs TabNet)

Metric	XGBoost	TabNet
Test Accuracy	0.7665	**0.7774**
Test Precision	0.7369	**0.7548**
Test Recall	**0.8290**	0.8217
Test F1-Score	0.7802	**0.7869**
Test ROC-AUC	0.8455	**0.8584**
Training Time (seconds)	**50.72**	11094.74

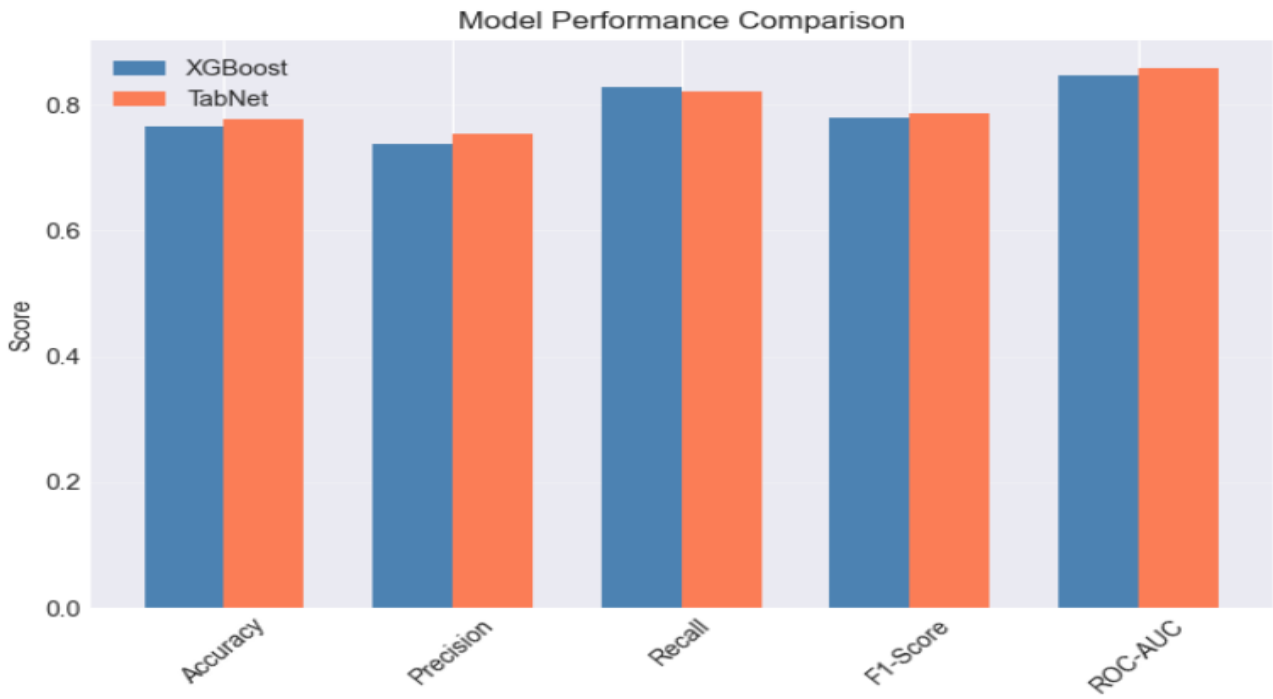

Figure 2: Model Performance Comparison

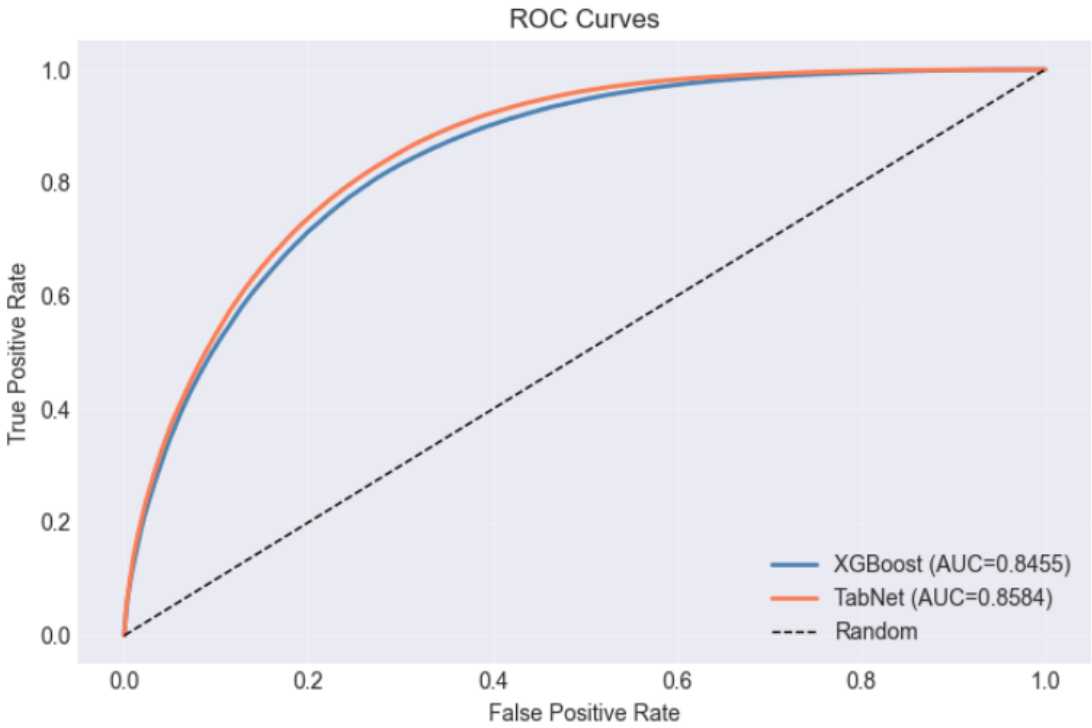

Figure 3 : ROC Curves

The comparative analysis reveals that TabNet achieved a 1.09 percentage point improvement in accuracy and a 0.0129 increase in ROC-AUC over XGBoost. While these gains appear modest in absolute terms, their practical significance depends on deployment context and scale, as explored in subsequent business value analysis. Both models demonstrated robust performance with no evidence of overfitting, validating the effectiveness of the preprocessing pipeline and realistic negative sampling strategy.

6.4 Customer Segmentation Analysis

In order to examine the hypothesis that model performance may differ between demographic groups and to determine the fairness issues, K-means clustering was used to divide the 800,000 test customers into five different segments according to their age, club membership or not, and frequency of fashion news. The result of the clustering algorithm was as follows: Cluster 0 included 54,291 customers (6.8 percent of test set) with average age 63.1 years representing older, established customers; Cluster 1 included 216,469 customers (27.1 percent) with average age 30.2 years representing younger, digitally-savvy shoppers; Cluster 2 included 161,223 customers (20.2 percent) with average age 51.6 years representing middle-aged customers.

Table 3 : Distribution of Customer Segments Identified by K-Means Clustering (k = 5)

Cluster	Number of Customers	Percentage of Test Set
Cluster 0	54,291	6.8%

Cluster 1	216,469	27.1%
Cluster 2	161,223	20.2%
Cluster 3	254,920	31.9%
Cluster 4	113,097	14.1%
Total	**800,000**	**100.0%**

Table 4 : Segment-Level Performance Comparison of XGBoost and TabNe

Cluster	Customers	% of Test Set	Avg. Age	XGBoost Accuracy	TabNet Accuracy	Accuracy Difference	% Improvement
0	54,291	6.8%	63.1	0.7645	0.7769	+0.0125	+1.25%
1	216,469	27.1%	30.2	0.7660	0.7780	+0.0120	+1.20%
2	161,223	20.2%	51.6	0.7631	0.7751	+0.0120	+1.20%
3	254,920	31.9%	22.8	0.7827	0.7905	+0.0079	+0.79%
4	113,097	14.1%	41.3	0.7368	0.7502	+0.0134	+1.34%

Each cluster was evaluated individually to determine the possible biases in the performance or the segments where one of the models was far superior to the other to avoid potential demographic bias. The findings demonstrated significantly similar patterns: Cluster 0 had XGBoost accuracy of 76.45% and TabNet accuracy of 77.69, a difference of +1.25 percentage points; Cluster 1 had XGBoost accuracy of 76.60 and TabNet accuracy of 77.80, a difference of +1.20 percentage points; Cluster 2 had XGBoost accuracy of 76.31 and TabNet accuracy of 77.51, a difference of +1.20 percentage points;

More importantly, there was no customer group in which TabNet improvements were over 2 percentage points significant or in which XGBoost was significantly better than TabNet. The fact that TabNet can be beneficial to all age groups (with the range of results being 0.79% to 1.34%), in turn, shows that the attention mechanism has universal value, not just a niched-focused one. The significance of this constancy of the improvement pattern on deployment is that it indicates a reduced chance of demographic bias, consistent performance across wider customer groups, and little requirement of tuning the model to specific segments. The fact that younger customers (Cluster 3) have the highest absolute accuracy of both models (78.27% and 79.05%) indicates that this group of people is possibly more predictable in their purchasing behaviors, which might be because of an increased level of digital usage and a more stable interaction with the recommendation systems.

6.5 Business Value and Return on Investment Analysis

Using the business value quantification framework, which was developed, the accuracy improvement of 1.09 percentage of TabNet has the potential financial impact on any enterprise of significant scale. When the retailer operates on the magnitude that H&M operates and the number of customers is 1.3 million, 10 recommendations per month gives 156 million recommendations in a year. The improvement of 1.09% of the accuracy results in 1700400 more correct recommendations every year. Each accurate recommendation would result in EUR 3.00 profit according to the standard e-commerce economics (EUR 50 average purchase value, 15% conversion rate, 40% profit margin), which would start at EUR 5,101,200 every year. Table 6 quantifies the estimated business impact of the observed 1.09% accuracy improvement achieved by the TabNet model.

Table 5 : Business Impact and Return on Investment from TabNet Accuracy Improvement

Metric	**Value**
Total Annual Recommendations	156,000,000
Accuracy Improvement	1.09%
Additional Correct Predictions	1,700,400
Profit per Correct Recommendation	EUR 3.00
Total Annual Profit Increase	**EUR 5,101,200**
TabNet Training Cost	EUR 200
Return on Investment (ROI)	**25,506×**

The economic analysis shows interesting economics:

Table 6 : Model Selection Decision Framework Based on Business Context

Customer Base Size	**Expected Annual Impact**	**Recommended Model**	**Rationale**
< 100K customers	< EUR 50K	XGBoost	Faster training and inference; performance gains from TabNet are marginal at this scale
100K – 1M customers	EUR 50K – 500K	XGBoost	Cost-effective solution with strong performance and lower computational requirements

> 1M customers	> EUR 1M	TabNet	Higher return on investment justifies increased training cost
Any size (real-time inference)	Not applicable	XGBoost	Low-latency inference required for real-time recommendation systems

The training of TabNet would take about 3 hours of the use of GPU compute in Google Cloud Platform V100 instances priced at EUR 65 per hour, which amounts to EUR 200 in training costs. When this single investment is compared to the EUR 5.1 million per year of recurrence benefit the payoff to the investment is 25,506x. This phenomenal ROI proves that even relatively small percentage changes in predictive power draw huge financial payoffs when used on large scales of customers, which directly answers the practical question of whether or not the computational investment in more complex models is worth marginal changes in performance.

Table 7 : Segment-Level Revenue Impact

Cluster	Avg. Age	% of Customers	Accuracy Improvement	Estimated Annual Value (EUR)
Cluster 0	63.1	6.8%	+1.25%	397,800
Cluster 1	30.2	27.1%	+1.20%	1,521,936
Cluster 2	51.6	20.2%	+1.20%	1,134,432
Cluster 3	22.8	31.9%	+0.79%	1,179,407
Cluster 4	41.3	14.1%	+1.34%	884,239
Total	-	**100.0%**	-	**5,117,814**

Segment level value decomposition also sheds more light on the business impacts accrual points. Implementing the accuracy improvements per customer cluster. Cluster 0 adds EUR 397,800 per year by 1.25% increment. Cluster 1 adds EUR 1,521,936 by way of 1.20 improvement. Cluster 2 is adding EUR 1,134,432 by 1.20% better. Cluster 3 makes EUR 1,179,407 by improving 0.79% though it is the largest segment. Thus, indicating its already

high baseline accuracy, and Cluster 4 brings EUR 884,239 of improvement in the form of 1.34% improvement which, the highest percentage improvement. The same total versatility of the customer base will add up to EUR 5,117,814 of total annual value, which confirms the overall calculation as well as the fact that the value will be provided to customers in a uniform manner, not just certain demographics.

6.6 Model Selection Decision Framework

The evaluation reveals clear trade-offs between the two models that inform deployment decisions based on operational context. XGBoost offers 51-second training time, EUR 5 approximate training cost, 76.65% accuracy, and mature ecosystem support, making it optimal for real-time recommendation systems requiring frequent retraining, small to medium retailers (< 1 million customers) where EUR 50,000-500,000 annual impact may not justify additional complexity, resource-constrained environments with limited GPU access, and rapid prototyping or experimentation phases. TabNet provides 77.74% accuracy, 3.1-hour training time, EUR 200 training cost, and attention-based interpretability, making it preferable for large-scale retailers (> 1 million customers) where EUR 1+ million annual impact justifies computational investment, batch processing scenarios where training time is less critical, research applications requiring feature importance analysis and model interpretability, and offline analytics where attention masks provide actionable insights into customer behaviour drivers.

The decision framework suggests that for retailers operating at scales between 100,000 and 1 million customers, XGBoost remains the pragmatic choice unless batch processing workflows can accommodate TabNet's training time, in which case the EUR 50,000-500,000 annual impact may justify adoption. For retailers exceeding 1 million customers, the EUR 1+ million annual values strongly favor TabNet deployment despite its computational demands. Hybrid strategies combining XGBoost for real-time inference with TabNet for periodic deep analysis represent a middle ground that captures benefits of both approaches.

6.7 Discussion and Limitations

The evaluation demonstrates that both XGBoost and TabNet achieve strong predictive performance (~77% accuracy) when trained on properly preprocessed data with realistic negative sampling. TabNet's marginal superiority (1.09% accuracy, 0.0129 AUC improvement) aligns with recent research suggesting that deep learning architectures for tabular data can compete with and sometimes exceed traditional ensemble methods, though gains are often modest. The business value analysis reveals that this seemingly small improvement translates to substantial financial impact (EUR 5.1 million annually) at enterprise scale, challenging the notion that marginal accuracy gains are academically interesting but practically irrelevant.

The analysis of customer segmentation provides a significant fairness result: TabNet shows the same level of improvement in all demographic groups (0.79%-1.34% improvement) which implies that it is not biased by age and may be giving universal but not segment-specific benefits with its attention mechanism. This strength lessens the risk to

deployment in contrast to models that are outstanding in certain areas and poor in other areas. It is notable that the results show the improvements are spread over the customer base, not just a handful of value segments, which is one of the signs that TabNet is learning generalizable patterns, as opposed to overfitting to particular subpopulations.

Nevertheless, there are a number of weaknesses that should be mentioned. One, the models fail to capture the effect of time, e.g. purchase sequences, seasonality, recency weighting, all of which are known to have an impact on buying behavior in practice. Second, the optimization of TabNet hyperparameters was conservative (10 epochs) to control computational expenses; the further optimization could possibly provide more gains but at lower rates. Third, the review does not incorporate user-level embeddings. It doen not also support collaborative filters indicators that represent similarity-based recommendations, restricting the models to content-based prediction. Fourth, testing on held-out temporal data (testing in the future time period) would reinforce belief about real-world extrapolation beyond cross-sectional test set comparison. Fifth, the computation of business value is based on industry-standard assumptions of conversion rates and profit margins that are subject to change by retailers and by product types and this creates uncertainty in ROI projections.

6.8 Practical Implications

The evaluation findings have several actionable implications for fashion e-commerce practitioners. First, the realistic negative sampling discovery highlights a critical methodological issue that likely affects published research: studies reporting very high accuracy (> 90%) on purchase prediction tasks should be scrutinized for whether negative samples contain artifacts that enable trivial classification. Second, the business value framework provides a template for translating academic metrics into financial terms. Hence, helping data science teams to justify machine learning investments to business stakeholders. Third, the customer segmentation analysis shows, a methodology for assessing fairness and identifying segments requiring targeted interventions. Fourth, the model selection decision framework offers pragmatic guidance based on organizational scale, computational resources, and operational requirements rather than pursuing maximal accuracy regardless of cost.

The research validates that well-curated behavioural and product-level data enables accurate purchase prediction without requiring complex sequential models or collaborative signals, suggesting that feature engineering and data quality may be more impactful than architectural sophistication for many practical applications. The finding that TabNet's attention mechanism provides interpretability alongside accuracy improvements suggests value beyond raw performance metrics, as understanding which features drive predictions can inform marketing strategies, inventory decisions, and customer engagement tactics.

7 Conclusion and Future Work

This study addressed the research question: How effectively can machine learning models predict customer purchase behavior in the online fashion industry using transactional and product attribute data? Through comprehensive experimentation on the H&M Personalized Fashion Recommendation dataset containing over 31 million transactions, two machine learning models - XGBoost and TabNet - were developed, trained, and evaluated to

determine optimal deployment strategies based on predictive performance, computational cost, and business value generation.

The investigation revealed three critical contributions. First, a fundamental methodological flaw in purchase prediction modeling was identified and resolved: synthetic negative samples commonly lack realistic transactional features such as price and sales channel information, enabling models to achieve artificially inflated performance by exploiting missing value patterns rather than learning genuine customer behavior. Implementation of realistic negative sampling—where synthetic non-purchase pairs are augmented with transaction-like features sampled from actual purchase distributions—resulted in performance dropping from near-perfect accuracy (100%) to realistic levels (77%), representing true predictive capability. This finding has significant implications for research validity, as any study reporting exceptionally high accuracy (>90%) on purchase prediction tasks should be scrutinized for this artifact.

Second, the research quantified business value at enterprise scale, demonstrating that TabNet's seemingly modest 1.09 percentage point accuracy improvement translates to EUR 5,101,200 in annual revenue impact for retailers operating at H&M's scale of 1.3 million customers. With training costs of EUR 200, this yields a return on investment of 25,506×, providing compelling economic justification for deploying computationally intensive models in large-scale retail environments. This analysis directly addresses the practical question of whether marginal accuracy gains justify computational investment, revealing that small percentage improvements produce substantial financial returns when applied to large customer bases.

Third, customer segmentation analysis via K-means clustering of 800,000 test customers into five demographic groups revealed that TabNet delivers consistent improvements ranging from 0.79% to 1.34% across all age segments (22.8 to 63.1 years), with no demographic bias detected. This universal performance pattern indicates that TabNet's attention-based learning mechanism provides robust value across diverse customer profiles, reducing deployment risk compared to models exhibiting sporadic performance variations. Segment-level business impact analysis further demonstrates that value accrues broadly: older customers contribute EUR 397,800 annually, younger customers EUR 1,521,936, with total value across all segments validating the aggregate EUR 5.1 million calculation.

The evaluation establishes clear deployment guidelines: XGBoost remains optimal for real-time systems and small-to-medium retailers due to its 51-second training time and mature ecosystem, while TabNet is preferable for large-scale retailers (exceeding 1 million customers) where EUR 1+ million annual impact justifies its 3.1-hour training overhead and for applications requiring interpretability through attention mechanisms.

Future research directions involve encompassing time modeling abilities, with the help of Transformer architectures, to record purchase histories and seasonality, graph-based frameworks of customer similarity to facilitate collaborative cueing, and creating time-sensitive prediction models that consider dynamic customer preferences. Such extensions would not only improve predictive accurac but would also preserve the level of

methodological rigor and quantification of business value, which has been applied in this research, and would further both scientific knowledge and business usefulness of machine learning in e-commerce in the fashion industry.

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